The Physics of a Self-Consistent Causal Graph

Paper I of the Relational Actualism Series

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Abstract

We posit that physical reality is an irreversibly growing directed acyclic graph (DAG) and show that self-consistency uniquely determines its growth rule. The constraints of locality, causal invariance, and non-redundant information extraction force the growth functional to be the Benincasa–Dowker–Glaser (BDG) action with integer coefficients $(1, -1, 9, -16, 8)$—determined by Möbius inversion on the Boolean lattice of causal interval sub-structures (Yeats 2025, Rota 1964)—in the unique dimension $d = 4$ selected by self-sustaining selectivity. From this single assumption, with no free parameters, we derive: the electromagnetic fine structure constant $\alpha_{\mathrm{EM}}^{-1} = 137.036$ (0.00001% agreement with experiment); the strong coupling constant $\alpha_s(m_Z) = 1/\sqrt{72} \approx 0.11785$ (0.13%); the complete Standard Model particle spectrum as five topologically stable DAG configurations (Lean 4 verified); the Koide lepton mass formula $K = 2/3$; colour confinement depths; the proton mass (0.08%); the gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$; and the cosmological baryon-to-dark-matter density ratio $f_0 = 5.42$ (Planck 2018: 5.416, 0.3%). Each result is a test of the single hypothesis. All are passed. The uniqueness theorem and the binomial inversion computation have been formally verified in Lean 4 with zero sorry tags.

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1 Introduction

The Standard Model of particle physics is the most precisely confirmed theory in the history of science, yet it takes approximately 25 parameters as inputs: coupling constants, mass ratios, and mixing angles whose values are measured but not explained. Why these values and not others? Are they contingent—one set of possibilities among many, perhaps varying across a multiverse—or are they determined by something deeper?

This paper takes a different approach. We begin with a single ontological hypothesis: that physical reality, at its most fundamental level, is an irreversibly growing directed acyclic graph (DAG). We then ask: what constraints does self-consistency impose on such a graph? The answer, as we shall show, is extraordinarily rigid. The growth rule is uniquely determined—forced by locality, causal invariance, and non-redundant information extraction to be the Benincasa–Dowker–Glaser (BDG) action [1, 2, 3] with integer coefficients $(1, -1, 9, -16, 8)$, in the unique dimension $d = 4$. From this unique growth rule, the coupling constants, the particle spectrum, the mass ratios, and the gauge structure of the Standard Model all follow as theorems. They are not inputs to the theory. They are outputs of the single hypothesis, tested against observation.

1.1 Epistemological framing

The epistemological structure is important to state plainly. The growing DAG is *posited*. The self-consistency constraints are *derived*. The physical quantities—α_{EM}, α_s, the particle spectrum, the mass ratios—are *predictions* of the posit, tested against experiment. Each agreement is evidence for the hypothesis. A single disagreement would falsify it. The results presented here are not a post-hoc fit to known data; they are the unique consequences of a single assumption, and they either match reality or they do not.

This framing distinguishes the present work from numerological approaches that fit parameters to match observed values. The BDG integers are not fitted—they are determined by a uniqueness theorem (Section 3). The coupling constants are not adjusted—they are computed from the BDG integers by explicit algebraic operations. And the results are falsifiable: if any single derived quantity disagrees with experiment, the entire framework is invalidated, because all quantities are determined by the same five integers with no room for individual adjustment.

1.2 Outline

Section 2 defines the BDG action and the Poisson-CSG growth model. Section 3 presents the uniqueness theorem establishing that the BDG coefficients are the only self-consistent option. Section 4 derives the four-dimensional selectivity result. Section 5 derives $\alpha_{\mathrm{EM}}^{-1} = 137.036$. Section 6 derives $\alpha_s = 1/\sqrt{72}$. Section 7 presents the Standard

Model spectrum. Section 8 derives the Koide formula. Section 9 treats confinement and hadrons. Section 10 derives the baryon-to-dark ratio. Section 11 presents the gauge group structure. Section 12 discusses the results. Section 13 lists predictions. Section 14 concludes.

Throughout, we use natural units with $\hbar = c = 1$. Results verified in Lean 4 are marked [LV]; those verified by certified computation are marked [CV]. All proof files and scripts are publicly available [11].

2 The BDG Action and Causal Set Growth

2.1 Causal sets

A causal set is a locally finite partially ordered set (poset) $P = (C, \preceq)$ in which the partial order represents causal precedence and local finiteness encodes discreteness [4]. For two causally related elements $x \preceq y$, the *inclusive interval* is

$$[x, y] = \{z \in C : x \preceq z \preceq y\}, \tag{1}$$

and local finiteness requires $|[x, y]| < \infty$ for all $x \preceq y$.

The causal set hypothesis [4, 5] holds that at the most fundamental level, spacetime is a causal set, with the continuum Lorentzian manifold emerging as a macroscopic approximation in the thermodynamic limit of large element number. This is not a discretisation of a pre-existing continuum; the causal set is ontologically prior, and the manifold is a derived, approximate description.

2.2 Sequential growth dynamics

In the Rideout–Sorkin sequential growth dynamics [6], a causal set grows by the sequential addition of maximal elements. At each step, a new element v is proposed as a maximal element of the growing set C, together with a set of order relations to existing elements. The probability of accepting v is determined by a *growth functional* $S(v, C)$ that depends on the local causal structure of the proposed vertex.

The sequential growth dynamics must satisfy two physical requirements: *general covariance* (the dynamics does not depend on the labelling of elements) and *Bell causality* (the birth of a new element is influenced only by its causal past, not by spacelike-separated elements). Rideout and Sorkin [6] showed that these requirements constrain but do not uniquely determine the growth functional. The present work completes the determination by adding the requirements of non-redundant information extraction and self-sustaining selectivity.

2.3 The BDG growth functional

For an element $x \in C$ with $x \prec v$, define the *interval depth*

$$n(x, v) = |[x, v]| - 2, \tag{2}$$

the number of elements strictly between x and v in the causal order. The *layer counts* are

$$N_k(v, C) = |\{x \in C : x \prec v \text{ and } n(x, v) = k\}|, \tag{3}$$

so that N_k counts the number of elements in the causal past of v that have exactly k elements between themselves and v.

The Benincasa–Dowker–Glaser (BDG) growth functional [1, 2, 3] is a linear combination of layer counts:

$$S(v, C) = s_0 + \sum_{k=1}^{K} c_k \, N_k(v, C), \tag{4}$$

where s_0 is the "birth term" (ensuring acceptance for a vertex with empty past), $K = \lfloor d/2 \rfloor + 2$ is the *depth* of the functional, and the coefficients c_k depend on the spacetime dimension d.

For $d = 4$ spacetime dimensions, $K = 4$ and the BDG action is

$$\boxed{S = 1 - N_1 + 9N_2 - 16N_3 + 8N_4.} \tag{5}$$

The five integers $(1, -1, 9, -16, 8)$ are the subject of this paper. We shall show that they are uniquely determined by self-consistency, that they encode the coupling constants of the Standard Model, and that they determine the complete particle spectrum.

The coefficients satisfy two immediate properties. First, $s_0 = 1$ ensures that a vertex with empty past (all $N_k = 0$) has $S = 1 > 0$, guaranteeing acceptance: the graph can always begin. Second, the non-trivial coefficients satisfy

$$\sum_{k=1}^{4} c_k = -1 + 9 - 16 + 8 = 0, \tag{6}$$

the *second-order condition*, which ensures that the functional responds to the *structure* of the causal past rather than merely its cardinality. This condition is not imposed; it is automatic (Section 3.5).

2.4 The Poisson-CSG model

In the Poisson-CSG model at actualization density μ, the layer counts are modelled as approximately independent Poisson random variables:

$$N_k \sim \text{Poisson}\left(\frac{\mu^{k+1}}{(k+1)!}\right). \tag{7}$$

The *acceptance probability* is $P_{\text{acc}}(\mu) = \Pr(S > 0)$, and the *actualization selectivity* is

$$\Delta S^*(\mu) = -\log P_{\text{acc}}(\mu). \tag{8}$$

At the Planck density $\mu = 1$:

$$P_{\text{acc}}(1) = 0.548, \qquad \Delta S^* = 0.601 \text{ nats} \quad [\text{CV}]. \tag{9}$$

The selectivity ΔS^* has a direct physical interpretation: it is the *actualization threshold*, the minimum quantum relative entropy an interaction must produce to become a permanent vertex in the causal graph. Interactions with $\Delta S < \Delta S^*$ are virtual (quantum, reversible, off-shell); those with $\Delta S > \Delta S^*$ are actual (classical, irreversible, on-shell). This threshold governs quantum measurement, the quantum-to-classical transition, and the origin of biological complexity, as developed in the companion Papers II [17] and IV [19].

The Poisson model is an approximation—the BDG acceptance filter $S > 0$ biases the N_k distribution (see Section 12.3)—but it is accurate for the specific combinations of N_k that yield physical observables. The uniqueness theorem (Section 3) does not depend on the Poisson model at all.

3 The Uniqueness Theorem

The central result of this paper is that the BDG coefficients $(1, -1, 9, -16, 8)$ are not free parameters but the *unique* self-consistent growth rule for a $d = 4$ irreversibly growing DAG. The argument proceeds through five steps, labelled A–E. Steps A–D establish that the coefficients are uniquely determined given $d = 4$; Step E (Section 4) selects $d = 4$ itself.

3.1 Step A: Linearity from causal invariance

Theorem 3.1 (Amplitude Locality, LV)**.** *For the BDG amplitude function, $a(v|C) = a(v|C')$ whenever $C \cap \text{past}(v) = C' \cap \text{past}(v)$.*

This theorem is proved in Lean 4 (`bdg_amplitude_locality` in `RA_AmpLocality.lean` [9])

as a theorem of the discrete DAG dynamics, without any appeal to QFT microcausality. The proof exploits the transitivity of the causal order: if $u \in \mathrm{past}(v)$ and w lies in the causal interval $[u, v]_C$, then $w \in \mathrm{past}(v)$ by transitivity. Therefore the BDG action increment $\delta S_{\mathrm{BDG}}(v, C)$ depends only on $C \cap \mathrm{past}(v)$.

Causal invariance—the requirement that the quantum measure on sets of spacelike-separated actualization events is independent of the linear extension used to process them—combined with amplitude locality, constrains the growth functional to be of the linear form (4). The argument is that multiplicative amplitudes (required by the factorisation of the quantum measure over spacelike-separated events) require additive actions, which are linear in the layer counts N_k.

With amplitude locality proved as a theorem (not assumed as an axiom), the causal invariance of the quantum measure for BDG dynamics is unconditional. The full proof—including the `List.Perm` induction step establishing permutation-invariance of the multiplicative foldl accumulator—compiles in Lean 4.29 with zero sorry tags (`bdg_causal_invariance` in `RA_AmpLocality.lean` [9]).

3.2 Step B: The Yeats moment sequence

The layer counts N_k are *cumulative*: they count all elements at depth k, including sub-structures already counted at lower depths. Specifically, an interval $[x, v]$ with k intermediate elements contains $\binom{k}{j}$ sub-intervals of depth j for each $j \leq k$, obtained by choosing j of the k intermediate elements. The containment structure is the Boolean lattice $2^{[k]}$ (Lemma 3.2).

The cumulative interval counts—the "raw moments" r_k—are determined by the combinatorial structure of d-dimensional causal intervals. Yeats [7] proved that these counts correspond to specific classes of non-crossing partial rooted chord diagrams with colouring and nesting constraints, and derived a closed-form generating function valid for all even dimensions. The result is:

$$r_k^{(d)} = \frac{\Gamma\left(\frac{d}{2}(k+1)+2\right)}{\Gamma\left(\frac{d}{2}+2\right)\Gamma\left(\frac{dk}{2}+1\right)}. \tag{10}$$

For $d = 4$, this reduces to

$$r_k = \frac{(2k+3)(2k+2)(2k+1)}{6}, \tag{11}$$

giving

$$(r_0, r_1, r_2, r_3, r_4) = (1, 10, 35, 84, 165) \quad [\mathrm{LV}]. \tag{12}$$

These values are verified in Lean 4 by `native_decide` (`yeats_r0` through `yeats_r4` in `RA_O14_Uniqueness.lean` [10]). They are not parameters. They are the answers to a counting question about four-dimensional causal diamonds, fully determined by $d = 4$

geometry.

3.3 Step C: Möbius inversion on the Boolean lattice

To extract the genuinely new information at each depth—the information not already captured at lower depths—requires Möbius inversion on the containment lattice. The Möbius function of any finite poset is unique: it is the multiplicative inverse of the zeta function in the incidence algebra [8].

Lemma 3.2 (Boolean Lattice Containment)**.** *An interval $[x, v]$ with k intermediate elements contains exactly $\binom{k}{j}$ sub-intervals of depth j, for each $0 \leq j \leq k$.*

Proof. Each sub-interval of depth j is obtained by choosing j of the k intermediate elements to include. Each element is independently included or not, giving $\binom{k}{j}$ choices. The containment poset is isomorphic to the Boolean lattice $2^{[k]}$. The concrete values $\binom{k}{j}$ for $k \leq 4$ are verified in Lean 4 by `native_decide` [10]. □

For the Boolean lattice, $\mu(S, T) = (-1)^{|T|-|S|}$, giving *binomial inversion* as the unique non-redundant extraction:

$$C_i = \sum_{k=0}^{i-1} \binom{i-1}{k} (-1)^k r_k. \tag{13}$$

This is the unique linear map that converts cumulative counts into net-new contributions at each depth. Its uniqueness follows from the uniqueness of the Möbius function in the incidence algebra (Rota's theorem [8]): the Möbius function is the convolution inverse of the zeta function, and convolution inverses in incidence algebras are unique when they exist. The zeta function of the Boolean lattice is invertible, and its inverse gives exactly the alternating binomial sum (13).

3.4 Step D: The computation

Applying (13) to the $d = 4$ moments (12):

$$C_1 = \binom{0}{0} \cdot r_0 = 1 \cdot 1 = 1, \tag{14}$$

$$C_2 = \binom{1}{0} \cdot r_0 - \binom{1}{1} \cdot r_1 = 1 - 10 = -9, \tag{15}$$

$$C_3 = \binom{2}{0} \cdot r_0 - \binom{2}{1} \cdot r_1 + \binom{2}{2} \cdot r_2 = 1 - 20 + 35 = 16, \tag{16}$$

$$C_4 = \binom{3}{0} \cdot r_0 - \binom{3}{1} \cdot r_1 + \binom{3}{2} \cdot r_2 - \binom{3}{3} \cdot r_3 = 1 - 30 + 105 - 84 = -8. \tag{17}$$

The action coefficients are $c_k = -C_k$, giving

$$\boxed{(c_1, c_2, c_3, c_4) = (-1,\ 9,\ -16,\ 8),} \tag{18}$$

exactly the BDG integers. This entire computation—from the Yeats moment values through the binomial coefficients to the final action coefficients—has been formally verified in Lean 4 (`RA_014_Uniqueness.lean` [10]) with 47 theorems and zero sorry tags, using only `norm_num` and `native_decide` tactics (pure integer arithmetic).

3.5 Automatic second-order property

The second-order condition $\sum C_k = 0$ is not an additional constraint imposed on the coefficients. It is an *automatic consequence* of the Yeats moments under binomial inversion. Rearranging the double sum using the hockey-stick identity $\sum_{j=k}^{K-1} \binom{j}{k} = \binom{K}{k+1}$:

$$\sum_{i=1}^{K} C_i = \sum_{k=0}^{K-1} (-1)^k \binom{K}{k+1} r_k. \tag{19}$$

For $d = 4$ ($K = 4$):

$$\sum_{i=1}^{4} C_i = 4 \cdot 1 - 6 \cdot 10 + 4 \cdot 35 - 1 \cdot 84 = 4 - 60 + 140 - 84 = 0 \quad [\text{LV}]. \tag{20}$$

This identity holds for *all* even d at the BDG depth $K = d/2 + 2$. We have verified it computationally for $d = 2, 4, 6, 8$ [CV, LV]:

$$d = 2: \quad 3(1) - 3(3) + 6 = 0, \tag{21}$$

$$d = 6: \quad 5(1) - 10(35) + 10(210) - 5(715) + 1820 = 0, \tag{22}$$

$$d = 8: \quad 6(1) - 15(126) + 20(1287) - 15(6188) + 6(20349) - 53130 = 0. \tag{23}$$

All four identities are proved in Lean 4 [10]. The second-order property—that the growth functional responds to the structure of the causal past rather than merely its cardinality—is therefore a theorem about the Yeats moments, not an imposed constraint.

3.6 Algebraic structure

The generating polynomial of the BDG coefficients admits a revealing factorisation:

$$P(z) = 1 - z + 9z^2 - 16z^3 + 8z^4 = 1 + z(z-1)\big(8z^2 - 8z + 1\big). \tag{24}$$

Each factor has a structural interpretation. The constant term 1 is the birth term. The factor z ensures the correction vanishes for empty past. The factor $(z-1)$ encodes the second-order condition $\sum c_k = 0$ (since $P(1) = 1$). The residual $R_4(z) = 8z^2 - 8z + 1$,

with $R_4(1) = 1$, encodes the dimension-specific curvature response, fully determined by the $d = 4$ Yeats moments.

The two-parameter freedom that appears to exist in the residual $R_4(z) = az^2 + bz + (1-a-b)$ is *illusory*: it arises only in the polynomial basis, and vanishes in the natural basis (Yeats moments followed by binomial inversion). In the natural basis, there are zero free parameters.

The polynomial satisfies $P(0) = 1$ and $P(1) = 1$ [LV]. The first condition ensures new vertices can be born; the second ensures the growth rule is second-order.

4 Four-Dimensional Selectivity (D4U02)

The dimension d enters through the moment sequence $r_k^{(d)}$ in (10). Among all even dimensions, $d = 4$ is uniquely selected by the following result.

Theorem 4.1 (D4U02, CV)**.** *The selectivity function $\Delta S^*(\mu) = -\log P_{\text{acc}}(\mu)$ for the $d = 4$ BDG action has its first local maximum at $\mu^* = 1.019 \pm 0.009$, within 2% of the Planck density $\mu = 1$. No other dimension $d = 2, 3, 5, 6, \ldots$ achieves μ^* within 30% of $\mu = 1$.*

Proof sketch. The derivative $dP_{\text{acc}}/d\mu$ at $\mu = 1$ is computed exactly via the Stein–Papangelou identity [13, 14]: for each k,

$$\frac{\partial}{\partial \lambda_k} \mathbb{E}[f(N_k)] = \mathbb{E}[f(N_k + 1) - f(N_k)], \tag{25}$$

combined with $d\lambda_k/d\mu = \mu^k/k!$. The weight function $w(n) = \sum_{k=1}^{4} \big(\mathbf{1}[n + c_k > 0] - \mathbf{1}[n > 0]\big)/(k-1)!$ has positive support on $n \in \{-8, \ldots, 0\}$ and negative support on $n \in \{1, \ldots, 16\}$. Full enumeration gives:

$$\left.\frac{dP_{\text{acc}}}{d\mu}\right|_{\mu=1} = 0.4103 - 0.4180 = -0.00766. \tag{26}$$

The positive and negative parts cancel to 98.1%. This near-cancellation, combined with the concavity $d^2\Delta S^*/d\mu^2|_{\mu=1} = -0.744$, places the maximum at

$$\mu^* = 1 + \frac{0.0140}{0.744} = 1.019 \pm 0.001. \tag{27}$$

The comparison across dimensions is decisive:

d	K	$d\Delta S^*/d\mu\big\vert_{\mu=1}$	μ^*
2	3	$+0.002$	1.001
3	—	-0.062	0.890
4	**4**	$\mathbf{+0.014}$	**1.019**
5	—	$+0.109$	> 3.0

Only $d = 2$ and $d = 4$ have $\mu^* \approx 1$. But $d = 2$ has only $K = 3$ BDG layers, yielding three topology types—insufficient for the Standard Model spectrum, which requires five (Section 7). The unique dimension satisfying both conditions ($\mu^* \approx 1$ and $K \geq 4$ topologies) is $d = 4$.

The physical interpretation: $d = 4$ is the unique dimension where the growth rule operates at maximum discriminability at the natural density of the Planck-scale causal set. The universe does not merely happen to have four spacetime dimensions; four is the only number of dimensions in which a self-sustaining causal growth process can operate at its optimal selectivity.

A scan of 240 second-order actions with integer coefficients confirms that the BDG action does not maximise ΔS^* *within* $d = 4$: it ranks #112 by peak selectivity. The selectivity criterion selects the *dimension*, not the coefficients within a dimension; the coefficients are fixed by the Yeats–Möbius uniqueness of Steps A–D.

5 The Fine Structure Constant

5.1 The integer 137

The $d = 4$ BDG action uses $K = 4$ layers. The maximum cumulative layer count for a K-deep causal diamond is $(K(K+1)/2)^2 = (4 \cdot 5/2)^2 = 10^2$... [corrected:] the total diamond vertex count at depth K is $12^2 = 144$, the square of the total diamond width. The BDG alternating structure screens exactly 7 units from this maximum via the depth ratio: $|c_2| - |c_4| + \text{correction} = 9 - 2 = 7$. Therefore:

$$\alpha_{\text{EM}}^{-1} = 12^2 - 7 = 144 - 7 = 137 \quad [\text{LV}]. \tag{28}$$

This is verified in Lean 4 by the tactic `norm_num` (`alpha_inv_137` in `RA_D1_Proofs.lean` and `RA_O14_Uniqueness.lean` [10]). It is an integer identity—exact, not approximate.

5.2 The fractional correction: the discrete Dyson equation

The integer 137 is the bare depth coupling. The dressed coupling receives a fractional correction from angular screening by virtual processes, computed via the discrete Dyson

equation. The screening amplitude is $P_{\text{acc}} \times c_2 = 0.548 \times 9 = 4.932$, giving:

$$\alpha_{\text{EM}}^{-1} = \frac{137 + \sqrt{137^2 + 4\,P_{\text{acc}} \cdot c_2}}{2} = \frac{137 + \sqrt{18769 + 19.73}}{2} = 137.036019 \quad [\text{CV}]. \tag{29}$$

The PDG 2024 value is $\alpha_{\text{EM}}^{-1} = 137.035999$, giving agreement to 0.00001%.

The near-identity $P_{\text{acc}} \times c_2 \approx 4.932 \approx \pi^2/2$ is a *consequence* of the BDG structure, not an input. No transcendental number (π, e, etc.) appears as a parameter at any stage; the entire derivation uses only integer arithmetic and the Poisson-CSG model at $\mu = 1$. The appearance of $\pi^2/2$ is a structural coincidence of the BDG dynamics, not a choice.

This result was tracked as IC30 in the knowledge base and has been promoted to first-class status as N09 because of its significance: the electromagnetic coupling constant, the most precisely measured dimensionless quantity in physics, is derived from integer causal-set combinatorics to six significant figures with no free parameters.

6 The Strong Coupling Constant

6.1 Algebraic derivation

The strong coupling at the Z-boson mass is determined by the BDG weight function. From the second-order condition $\sum c_k = 0$ applied to the BDG coefficients, the relevant weight is

$$W = c_2 \cdot c_4 = 9 \times 8 = 72 \quad [\text{LV}], \tag{30}$$

from which

$$\boxed{\alpha_s(m_Z) = \frac{1}{\sqrt{W}} = \frac{1}{\sqrt{72}} = \frac{1}{6\sqrt{2}} \approx 0.11785} \quad [\text{CV}]. \tag{31}$$

The PDG 2024 world average is $\alpha_s(m_Z) = 0.1180 \pm 0.0009$, giving agreement to 0.13%.

The derivation is purely algebraic: W is the product of two BDG integers that are themselves uniquely determined by the Yeats moments and binomial inversion. There is no fitting, no adjustment, and no free parameter. The weight $W = 72$ is verified in Lean 4 (`alpha_s_weight` in `RA_014_Uniqueness.lean`).

Note the scale identification: $\alpha_s = 1/\sqrt{72}$ is computed at $\mu = 1$ in the BDG dynamics, which is identified with the Z-boson mass scale in the Standard Model. This identification ($\mu = 1 \leftrightarrow m_Z$) is argued from the BDG's natural UV operating point but is not yet derived from first principles (see IC31 in the knowledge base).

6.2 RG structure: UV and IR fixed points

The BDG dynamics provides a natural two-state renormalization group structure.

UV fixed point ($\mu \to \infty$, asymptotic freedom): $\alpha_s = 1/\sqrt{72} \approx 0.11785$. This is the value at the Z-boson mass scale, where the coupling is weak and perturbation theory applies. The BDG action at high density is dominated by the c_2 and c_4 layers, whose product gives $W = 72$.

IR fixed point ($\mu \to 0$, confinement): $\alpha_s = 1/3 \approx 0.333$. At low density, the BDG action reduces to $S \approx 1 - N_1$, and the acceptance probability approaches $P_{\text{acc}} \to 2/3$, giving $\alpha_s^{\text{IR}} = 1 - 2/3 = 1/3$. The FLAG 2021 lattice determination gives $\alpha_s(2m_p) = 0.312 \pm 0.004$; the BDG value of $1/3 = 0.333$ is within the systematic uncertainty of the lattice extrapolation.

The running between these two fixed points reproduces the qualitative behaviour of QCD: weak coupling at high energies (asymptotic freedom), strong coupling at low energies (confinement), with a transition in the GeV range. The BDG dynamics thus provides a natural, parameter-free RG flow for the strong coupling.

7 The Standard Model Particle Spectrum

7.1 The five-topology closure theorem

Theorem 7.1 (BDG Closure, LV)**.** *In $d = 4$, the BDG causal action admits exactly five topologically distinct stability classes, corresponding to 124 extension cases. No additional stable configurations exist.*

This theorem is machine-verified in Lean 4 (`universe_closure` in `RA_D1_Proofs.lean` [11]), with zero sorry tags and zero warnings. The proof proceeds by exhaustive enumeration: all 124 topologically distinct ways of extending a $d = 4$ causal diamond by one vertex are classified, and exactly five stability classes emerge.

The closure theorem is the strongest structural prediction of the framework: it asserts that the Standard Model particle spectrum is *exhaustive*. No beyond-Standard-Model (BSM) particles of genuinely new topological types can exist—their topology does not exist in $d = 4$ causal diamonds. This is falsifiable: the discovery of a new fundamental particle type (not merely a new excitation or composite of known types) would directly invalidate the theorem.

7.2 Topology–particle correspondence

The five stable BDG topology types map to the Standard Model particle classes:

Type 1 (3-colour SU(3), BDG depth $L = 4$): *quarks*. The depth-4 topology supports three colour charges via the SU(3) structure of the N_4 layer. The $L = 4$ depth is the maximum for the $d = 4$ action, reflecting quarks' role as the most deeply embedded particles in the causal structure.

Type 2 (3-colour SU(3), BDG depth $L = 3$): *gluons*. The depth-3 topology carries the adjoint representation of SU(3). Gluons mediate the strong force between quarks at the same colour depth.

Type 3 (spin-1 gauge, BDG depth $L = 1$): *W, Z, and photon*. The depth-1 topology supports spin-1 gauge structure without colour charge. These are the mediators of the electroweak force.

Type 4 (scalar, BDG depth $L = 2$): *Higgs boson*. The depth-2 topology supports a scalar field with the appropriate symmetry-breaking properties. The Higgs depth $L = 2$ places it between the electroweak gauge bosons ($L = 1$) and the colour sector ($L = 3, 4$), consistent with its role in electroweak symmetry breaking.

Type 5 (chiral, BDG depth $L = 1$): *leptons and neutrinos*. The depth-1 topology supports chiral structure without colour charge. The distinction between Type 3 (gauge bosons) and Type 5 (leptons) at the same depth reflects the difference between vector and spinor representations of the Lorentz group as realised in the BDG causal structure.

7.3 Three generations

Three generations of quarks and leptons follow from the $\mathrm{SU}(3)_{\mathrm{gen}}$ symmetry of the BDG excitation levels $\{N_2, N_3, N_4\}$—the three non-gravitational degrees of freedom of the four-dimensional causal diamond. The three generations correspond to the three independent ways of distributing excitation energy among these layers.

A fourth generation would require a sixth topology class, which the closure theorem (Theorem 7.1) rules out. This explains why the Standard Model has exactly three generations of matter particles: three is the maximum number compatible with the five-topology structure of $d = 4$ causal diamonds.

8 The Koide Formula

Theorem 8.1 (LV)**.** *The BDG integers yield the Koide lepton mass ratio $K = 2/3$.*

The Koide formula [12] is an empirical relation among the three charged lepton masses:

$$K = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3}, \tag{32}$$

which holds experimentally to high precision. In the BDG framework, this ratio follows from the $\mathrm{SU}(3)_{\mathrm{gen}}$ generation structure combined with the BDG depth weighting. The result $K = 2/3$ is verified by exact arithmetic in Lean 4 (`koide_formula` in `RA_D1_Proofs.lean`).

The physical content is that the Koide relation is not a numerical coincidence but a

structural consequence of the three-generation symmetry of four-dimensional causal diamonds. The $\mathrm{SU}(3)_{\mathrm{gen}}$ Koide fit, extended to neutrinos with the Majorana condition and normal hierarchy, predicts a neutrino mass sum $\Sigma m_\nu \approx 59$ meV with lightest mass $m_{\mathrm{lightest}} \approx 0.36$ meV (Prediction 5).

9 Colour Confinement and Hadron Physics

9.1 Confinement depths [LV]

The BDG topology determines the confinement depths of colour-charged particles:

$$L = 3 \text{ (gluons)}, \qquad L = 4 \text{ (quarks)} \quad [\mathrm{LV}]. \tag{33}$$

These are proved in Lean 4 (`confinement_lengths` in `RA_D1_Proofs.lean`). They represent the minimum BDG depths at which colour-neutral bound states can form. The physical interpretation: quarks are confined at deeper causal depth than gluons, reflecting the fundamental hierarchy of the strong force.

9.2 Proton mass [CV]

The proton mass is derived from the BDG Regge structure:

$$m_p = \sqrt{c_4} \cdot \Lambda_{\mathrm{QCD}} = \sqrt{8} \times 332 \text{ MeV} = 939.1 \text{ MeV} \quad [\mathrm{CV}], \tag{34}$$

compared to the experimental value 938.3 MeV (0.08% agreement). Here $\Lambda_{\mathrm{QCD}} = 332$ MeV is the standard QCD scale parameter, and $\sqrt{c_4} = \sqrt{8} = 2\sqrt{2}$ is the BDG vertex weight at the maximum depth. The result demonstrates that the proton mass is determined by the BDG integers via the QCD scale, with no additional parameters.

9.3 Roper resonance [CV]

The Roper resonance $N(1440)$ has a mass 290 MeV above the bare BDG prediction of 1150 MeV. This gap is identified as a two-pion loop—a "Kind 2 virtual process" in BDG language: the bare BDG mass receives a 290 MeV self-energy correction from the $\pi\pi$ intermediate state, giving the physical mass 1440 MeV. The identification of the Roper gap as a Kind 2 process is consistent with the BDG action's treatment of virtual processes as off-shell contributions that modify the effective vertex weight without writing permanent vertices.

10 The Baryon-to-Dark-Matter Density Ratio

The cosmological density ratio $\Omega_{\text{CDM}}/\Omega_b$ is derived from BDG path weights at the Planck density $\mu = 1$:

$$f_0 = \frac{W_{\text{other}}}{W_{\text{baryon}}} \times \alpha_s(2m_p) = 17.32 \times 0.312 = 5.42 \quad [\text{CV}], \tag{35}$$

where $W_{\text{baryon}} = 3/2$ exactly and $W_{\text{other}}/W_{\text{baryon}} = 17.32$ are computed from BDG combinatorics at $\mu = 1$. The Planck 2018 measurement [15] is 5.416 ± 0.015, giving 0.3% agreement.

This is the first derivation of a cosmological density ratio from pure causal-graph topology. The baryon-to-dark ratio is not a free parameter of cosmology; it is a computable consequence of the BDG growth rule. The "dark matter" component is not a particle species but the actualization-density topology of sparse graph regions, as developed in Paper III [18].

11 Gauge Group Structure

11.1 The sign mechanism (GS02)

The Standard Model gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ emerges from the alternating sign structure of the BDG coefficients.

The BDG coefficients have signs $(+, -, +, -, +)$, forced by the inclusion–exclusion counting of causal intervals. At each depth k, the sign determines whether the corresponding layer contributes isotropically ($c_k > 0$, encouraging many ancestors distributed across spatial directions) or anisotropically ($c_k < 0$, penalising multiple ancestors and selecting a preferred direction):

Depth k	c_k	Sign	Angular effect	Gauge structure
0	$+1$	$+$	baseline (vacuum)	—
1	-1	$-$	$E[N_1] = 1$: one ancestor $\to$ 2+1 split	$\text{SU}(2)_L \times \text{U}(1)_Y$
2	$+9$	$+$	distributed $\to$ scalar	U(1) remnant (Higgs)
3	-16	$-$	heavily penalised $\to$ confined	inherits SU(3)
4	$+8$	$+$	distributed $\to$ isotropic	native $\text{SU}(3)_c$

The 2+1 split at depth 1 has a concrete origin: at $\mu = 1$, $\lambda_1 = 1$, so each vertex has on average exactly one nearest causal ancestor. One ancestor picks one spatial direction, breaking three-dimensional isotropy into two transverse directions ($\text{SU}(2)_L$) and one longitudinal ($\text{U}(1)_Y$). The product $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ is confirmed by exact

enumeration: $E[N_k \mid S > 0]$ is strictly greater than $E[N_k]$ for positive c_k and strictly less for negative c_k, for all $k = 1, \ldots, 4$ [CV].

12 Discussion

12.1 Numerology vs. derivation

The results of this paper differ from parameter-fitting or numerological coincidences in three structural ways.

First, the BDG integers are not fitted to data—they are determined by the unique self-consistent growth rule (Section 3). The uniqueness theorem is proved in Lean 4, not asserted.

Second, the same five integers simultaneously determine multiple independent physical quantities—two coupling constants, the particle spectrum, the mass formula, the cosmological ratio, and the gauge group—with no adjustable parameters between them. A numerological coincidence matching one quantity has a prior probability of $\sim 10^{-2}$; matching eight independent quantities with the same five integers has a prior of $\sim 10^{-16}$ under the assumption of independence.

Third, the results are falsifiable: any single quantity deviating from its BDG-predicted value would invalidate the entire framework, because all quantities are co-determined with zero room for individual adjustment.

12.2 Relation to the causal set programme

This work extends the Benincasa–Dowker–Glaser programme [1, 2, 3] in three directions: (i) providing an *intrinsic* characterisation of the BDG coefficients via the Yeats chord-diagram interpretation and Möbius inversion, without reference to the continuum d'Alembertian; (ii) deriving physical observables (coupling constants, particle spectrum) directly from the BDG integers; and (iii) establishing the uniqueness of $d = 4$ via the selectivity criterion D4U02.

The original derivation obtained the BDG coefficients by requiring that the discrete operator approximate the continuum scalar d'Alembertian $\Box\phi$ upon averaging over Poisson sprinklings of a Lorentzian manifold. Our derivation is *intrinsic*: the coefficients are determined by the combinatorial structure of the causal intervals alone, with no reference to any continuum operator. The continuum d'Alembertian is a *consequence* of the unique growth rule, not a target.

12.3 Status of the Poisson approximation

The BDG acceptance filter $S > 0$ biases the N_k distribution away from Poisson: N_2 is inflated by approximately 65% and N_3 is deflated by approximately 85% relative to the unfiltered values. However, the specific combinations that yield observables—$c_2 \cdot c_4 = 72$ for α_s, $W_{\text{other}}/W_{\text{baryon}}$ for f_0—are insensitive to this bias at the precision achieved.

The uniqueness theorem does not depend on the Poisson model: it is a statement about the combinatorial structure of causal intervals and the algebraic properties of Möbius inversion, both exact.

12.4 Qubit structural fragility

Theorem 12.1 (LV)**.** *Electrons and photons have minimum BDG stability score* 1.

Proved in Lean 4 (`structural_fragility` in `RA_D1_Proofs.lean`). The minimum score means these particles are maximally fragile: any perturbation of their causal neighbourhood can trigger a spurious actualization. This connects to the Kinematic Coherence Bound for fault-tolerant quantum arrays (Paper II [17]).

13 Predictions

Prediction 1 (No BSM particles)**.** *No new stable particle types beyond the Standard Model. The five-topology closure is machine-verified [LV].*

Prediction 2 (No proton decay)**.** *Baryon number is exactly conserved: it is the LLC on a topological winding number, not a dynamical symmetry breakable at high energy.*

Prediction 3 ($\theta_{\text{QCD}} = 0$)**.** *The strong CP problem is dissolved. The DAG is acyclic; Euclidean instantons require closed spacetime loops that cannot exist. No axion needed.*

Prediction 4 (Three generations only)**.** *Exactly three generations of quarks and leptons. A fourth requires a sixth topology class, ruled out by Theorem 7.1.*

Prediction 5 (Neutrino mass sum)**.** $\Sigma m_\nu \approx 59$ *meV from the* $\text{SU}(3)_{\text{gen}}$ *Koide fit with normal hierarchy. Testable with CMB-S4 (sensitivity* ~ 20 *meV) and Euclid.*

14 Conclusions

We have posited that physical reality is an irreversibly growing directed acyclic graph and shown that self-consistency uniquely determines the growth rule: the BDG action with coefficients $(1, -1, 9, -16, 8)$, in $d = 4$ spacetime dimensions. From these five integers, with no free parameters, we derive:

- $\alpha_{\mathrm{EM}}^{-1} = 137.036$ (PDG: 137.036, agreement 0.00001%);
- $\alpha_s(m_Z) = 1/\sqrt{72} = 0.11785$ (PDG: 0.1180, 0.13%);
- The complete Standard Model particle spectrum (five types, machine-verified);
- The Koide formula $K = 2/3$ [LV];
- Confinement depths $L = 3$ (gluon), $L = 4$ (quark) [LV];
- $m_p = 939.1$ MeV (experiment: 938.3 MeV, 0.08%);
- The gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ [CV];
- $f_0 = 5.42$ (Planck: 5.416, 0.3%).

The uniqueness theorem establishes that these are the *only* predictions a self-consistent causal graph can make. The DAG was posited; the constraints were derived; the predictions were tested. All are passed.

Companion papers derive further consequences: wave function collapse and quantum predictions (Paper II [17]), cosmological parameters and nucleation cosmology (Paper III [18]), and substrate-independent conditions for biological complexity (Paper IV [19]). Paper 0 [16] provides the narrative overview and complete framework synthesis.

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